



Shri Vile Parle Kelavani Mandal's

DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING

(Autonomous College Affiliated to the University of Mumbai)

NAAC Accredited with "A" Grade (CGPA : 3.18)



Innovative Product Development Report

on

Automated Gear Defect Inspection System Using Raspberry Pi and PLC

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Department of Electronics and Telecommunication Engineering

This is to certify that

"Automated Gear Defect Inspection System Using Raspberry Pi and PLC"

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Students of Electronics and Telecommunication Engineering have successfully completed their Innovative Product Development-IV required for the fulfillment of SEM VI during the first half of the year 2025. The project report has been assessed and found to be satisfactory.

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Abstract

This paper describes the design and implementation of an Automated Gear Defect Inspection System that uses visual inspection powered by machine learning and a servo motor-driven pushing mechanism for online sorting of PLA gears on a conveyor belt. The system employs a Raspberry Pi 5 board with the YOLOv8 machine learning model to classify gears into defective or non-defective categories based on real-time camera input. Upon detection of a defective gear, a linear actuator equipped with a rack-and-pinion mechanism driven by an MG995 servo motor ejects the faulty component off the conveyor belt. The conveyor belt measures 700 x 100 mm with a total circumferential length of 1310 mm and is driven by a 12V DC motor. The system achieves above 91% detection accuracy at a throughput of approximately 12 gears per minute.

Keywords: *gear inspection, YOLO, Raspberry Pi, PLA, conveyor sorting, defect detection, machine learning*

Introduction

Object detection through automated visual inspection has become the backbone of quality assurance in manufacturing. Manual inspection depends on human operators who are prone to errors due to fatigue and inconsistency. Recent developments in edge computing and deep learning have opened the possibility of deploying real-time object detection models on affordable single-board computers (SBCs) at manufacturing sites.

Gears are critical mechanical components that affect the overall efficiency and safety of any machine system. Flaws such as missing teeth, cracks, pits, and dimensional deviations can lead to early failure. Manual inspection of gears is a time-consuming process, particularly when large quantities are produced using Fused Deposition Modeling (FDM).

This paper introduces a gear defect inspection system comprising three integrated modules: (1) a motor-driven conveyor belt for gear transport, (2) a Raspberry Pi 5 executing a YOLOv8 algorithm for defect classification, and (3) a linear actuator based on rack-and-pinion driven by an MG995 servo motor for physical separation of defective gears. The system operates without human involvement after initialization and delivers consistent, accurate classification at high throughput.

Three key innovations of this project include: (i) a fine-tuned YOLOv8-nano network for binary gear classification on Raspberry Pi 5.

(ii) a cost-effective and compact mechanical sorting device based on rack-and-pinion and (iii) a fully functional working prototype capable of real-time sorting.

Literature Review

Automated visual inspection has been widely adopted in quality control processes within modern manufacturing to improve productivity and product quality [1]. Conventional inspection approaches involve human operators, which introduce disadvantages related to human error and fatigue.

Gear components are crucial mechanical elements whose malfunction adversely affects machine operation and increases maintenance costs. Research confirms that gear-related defects account for a significant proportion of mechanical failures and operational downtime [2]. Common inspection techniques such as manual examination and laboratory analysis are inefficient and unsuitable for real-time use.

Advances in artificial intelligence and edge computing now make it feasible to deploy object detection neural network models directly on inexpensive edge devices. Deep learning models like YOLOv8 have demonstrated high effectiveness in defect detection and can be practically deployed in industrial settings [3]. Furthermore, industrial automation systems increasingly employ Programmable Logic Controllers (PLCs) and intelligent communication devices for real-time control and decision-making in manufacturing [4]. Additional relevant works include PLC-based color object sorting using image processing [5], automated fruit inspection systems combining OpenCV and PLC for Industry 4.0 [6], low-contrast surface defect detection using improved YOLOv7 [7], and deep learning methods for substation equipment defect detection [8]. Machine vision-based automatic sorting [9] and AI-driven manufacturing quality control [10] further validate the feasibility of the proposed approach.



Methodology / Theory

Working Principle

The system is built around four interconnected subsystems: the mechanical conveyor subsystem, the vision and inference subsystem, the control and actuation subsystem, and the power management component. Gears are placed at the inlet of the conveyor and transported through the inspection area. Non-defective gears continue to the outlet bin, while defective gears are laterally ejected by the rack-and-pinion actuator. The overall workflow is as follows:

1. Gear is placed on the conveyor belt and transported at a controlled speed (approximately one gear every 5 seconds).
2. The USB camera (1080p, 30fps) captures a live video stream and frames are acquired via OpenCV VideoCapture on Raspberry Pi 5.
3. Frames are resized to 640x640 and pixel values normalized to [0,1] before being passed through the YOLOv8-nano model.
4. The model outputs a bounding box and class confidence score. If the defect confidence exceeds 0.75, a GPIO command is issued.
5. A GPIO HIGH signal triggers the MG995 servo motor via PWM pulse, rotating 90 degrees in 150 ms and producing 80 mm lateral movement through the rack-and-pinion mechanism, ejecting the defective gear.
6. The servo resets to its original position, and the system is ready for the next gear.

Hardware Components

Component	Specification	Purpose
Raspberry Pi 5	Quad-core Cortex-A76, 4GB RAM	Image capture & YOLO inference
MG995 Servo	180°, 10 kg·cm torque	Push actuator arm control
12V DC Motor	300 RPM, 5A rated	Conveyor belt drive
Conveyor Belt	700×100 mm, 1310 mm circumference	Gear transport
Rack & Pinion	Linear stroke 80 mm	Defect gear ejection
PLA Gears	FDM printed, ±0.2 mm tolerance	Test subjects (pass/fail)
USB Camera	1080p, 30fps	Real-time gear imaging

Dataset and Model Training

The dataset was collected in a controlled environment with a dedicated lighting setup at the inspection station. Data labeling was performed manually using Roboflow, and augmentation was applied via rotation, brightness jitter, and Gaussian noise. A total of 1,000 images of PLA gears were used — 500 defective and 500 non-defective.

The YOLOv8-nano model was trained for 100 epochs on a cloud GPU (NVIDIA T4) using Ultralytics with the following hyperparameters: learning rate = 0.01, batch size = 16, image size = 640, and IoU threshold = 0.5.



Block Diagram / System Architecture / Workflow

The complete system architecture integrates five major functional blocks:

- (1) Mechanical Input System — conveyor belt with DC motor and PLC;
- (2) Image Acquisition System — USB camera with controlled capture, lighting, and stabilization;
- (3) Processing Unit (Raspberry Pi) — frame preprocessing and YOLOv8 model inference;
- (4) AI Model (YOLO-Based Detection) — classifying each gear frame and computing confidence score;
- (5) Control System (Arduino/PLC) — position synchronization and actuation delay;
- (6) Actuation System — servo motor rotating to specific angle, rack-and-pinion mechanism;
- (7) Output System — defective gears ejected sideways, non-defective gears forwarded on conveyor; and
- (8) Decision Logic Layer — determining gear status and triggering the rejection signal.

Results

Detection Performance

The YOLOv8-nano model, after training, achieved an mAP@0.5 of 0.946 on an unseen test dataset of 200 images. The table below summarizes object detection performance:

Metric	Defective Class	Non-Defective Class
Precision	0.872	0.978
Recall	0.960	1.000
F1-Score	0.914	0.989
mAP@0.5	0.946	0.995

System Throughput

Latency from frame capture to servo actuation was assessed over 100 cycles. The average total latency is 312 ms, broken down as: frame capture = 33 ms, inference = 218 ms, GPIO/servo operation = 61 ms. Given the 5-second interval between gears on the conveyor, the system handles 12 gears per minute without any detection failures.

Sorting Accuracy

Sorting performance was evaluated on 300 gears (150 defective and 150 non-defective). Out of 300 gears, 276 were correctly identified and sorted, yielding an overall sorting accuracy of 92%. Errors were primarily caused by partial occlusion of gears at conveyor edges and inconsistent lighting conditions.

Power Consumption

The total power consumption of the system is approximately 14.3 W, comprising 6.2 W for the Raspberry Pi 5, 3.8 W for the servo motor, and 4.3 W for the motor driver chip. The system can be powered by a standard 5V/5A supply for the Raspberry Pi and a 12V/2A supply for the motor driver.



Future Scope

The current system demonstrates a strong proof of concept for automated gear defect detection. Several enhancements are planned for future iterations:

- Multi-class defect classification to distinguish between specific fault types such as tooth fracture, surface cracks, and dimensional deviations.
- Integration of a Programmable Logic Controller (PLC) to provide enhanced industrial-level control, safety interlocks, and fail-safe mechanisms.
- Development of a Human Machine Interface (HMI) for real-time system operation, parameter adjustment, and status monitoring.
- Implementation of a web-based dashboard for traceability, inspection data logging, and analytics.
- Expansion of the training dataset with gears printed using PETG and ABS materials at varying infill densities to improve model generalization.
- Improved lighting systems and camera positioning to reduce sorting errors caused by occlusion and uneven illumination.

Budget Sheet

Sr.	Component	Specification	Qty	Cost (₹)
1	Raspberry Pi 5	4GB RAM	1	~7,500
2	MG995 Servo Motor	180°, 10 kg·cm	1	~350
3	12V DC Motor	300 RPM, 5A	1	~600
4	USB Camera	1080p, 30fps	1	~800
5	Conveyor Belt & Frame	700×100 mm, 3D printed	1	~3,000
6	Rack & Pinion Mechanism	80 mm stroke, 3D printed	1	~500
7	PLA Filament	FDM grade, ±0.2 mm tol.	2 rolls	~1,500
8	Motor Driver & Electronics	PWM, GPIO, wiring	1 set	~800
9	Power Supply	5V/5A + 12V/2A	1 set	~900
10	Miscellaneous	Connectors, fasteners, etc.	-	~500
			Total	~16,450

Note: The total hardware cost of approximately ₹16,450 – ₹20,000 positions this system as a highly cost-effective alternative to traditional industrial vision inspection machines.



Applications

The Automated Gear Defect Inspection System has broad applicability across multiple industrial and research domains:

- Manufacturing Quality Control: Real-time inline inspection of mechanical components (gears, bearings, pulleys) on production lines to eliminate defective parts before assembly.
- Automotive Industry: Automated inspection of gear components in transmission and drivetrain assemblies where precision tolerances are critical.
- 3D Printing / Additive Manufacturing: Post-print quality verification for FDM-printed mechanical parts to ensure dimensional accuracy and structural integrity.
- Small and Medium Enterprises (SMEs): Low-cost, deployable quality inspection solution suitable for workshops and small-scale manufacturers who cannot afford industrial vision systems.
- Industry 4.0 and Smart Manufacturing: Integration with IoT platforms, SCADA systems, and digital twins for real-time production monitoring, traceability, and predictive maintenance.
- General Discrete Part Sorting: The modular architecture of the system allows adaptation for sorting other small discrete components beyond gears, such as fasteners, caps, or electronic components.

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